



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO RAFI FUELS

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

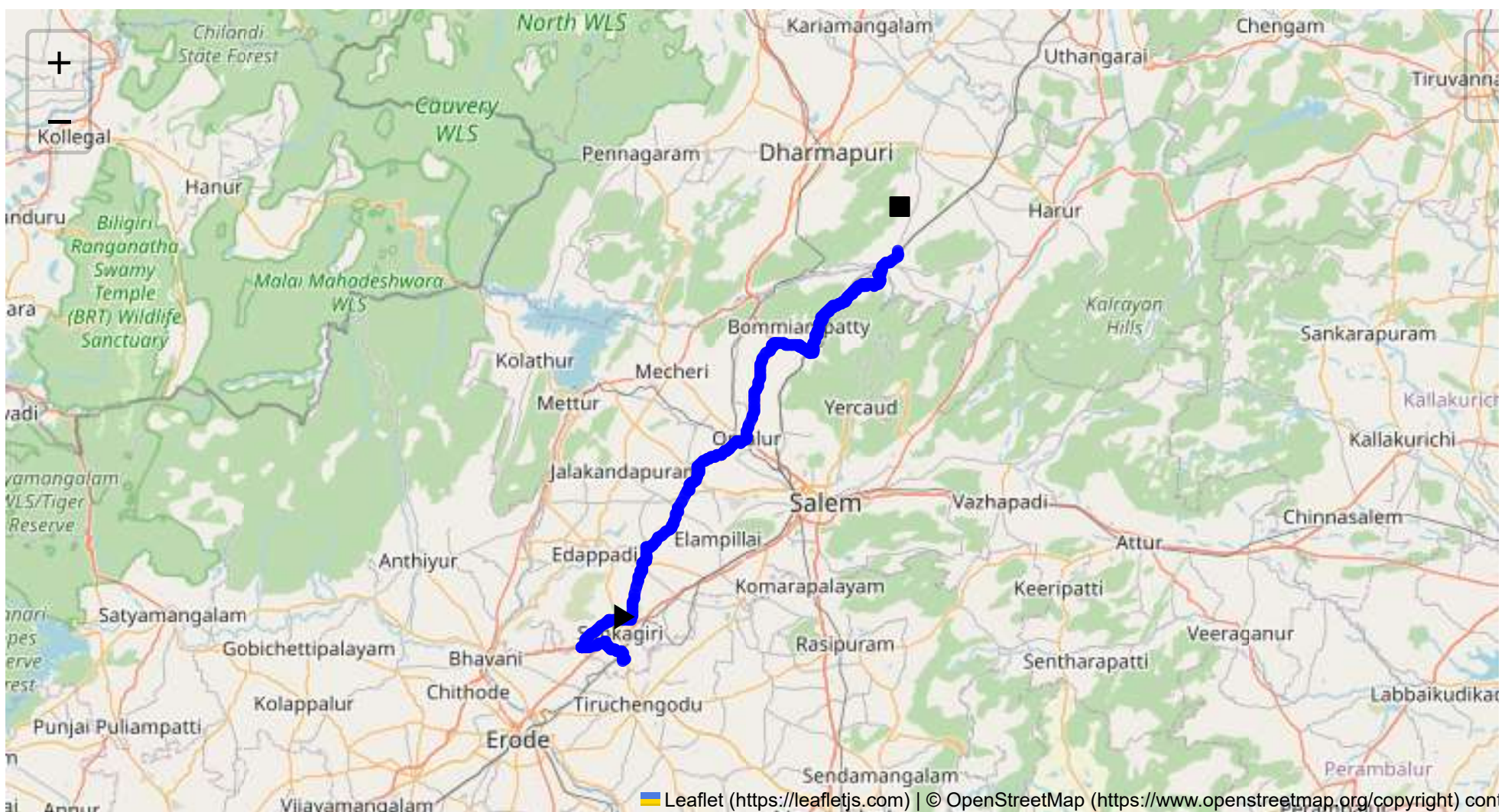
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 99.35 km**  
**Estimated Duration: 2.4 hours**  
**Adjusted Duration (Heavy Vehicle): 3.0 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.99874, 78.258388)**



## Welcome to the Journey Risk Management Study

Route Safety Analysis for Heavy Vehicle Transport from Sangagiri to Gandhinagar via Designated Points in Tamil Nadu

### 1. Overview of the Route Map:

- The route covers approximately 99.35 kilometers, passing through Sangagiri, Puthur, Padaiveedu, Sangagiri (again via another point), Erumaipatti, Desavilakku, Omalur, Danishpet, and K.N.Pudur,



before reaching Gandhinagar.

- Roads utilized include Salem - Kochi Highway, Omalur-Sankagiri Road, and Mettur Main Road (SH 86), all of which are major routes linking small towns and industrial areas.

## **2. Typical Weather Conditions and Weather-Related Hazards:**

- Tamil Nadu generally experiences three climatic conditions: summer (April-June), monsoon (July-September and October-November), and winter (December-March).
- Potential weather-related hazards include heavy rainfall and subsequent flooding during the monsoon seasons, which can affect visibility, road conditions, and potentially cause landslides in hilly areas near Danishpet.

## **3. Traffic Patterns and Congestion-Prone Areas:**

- Expect peak traffic hours in the mornings (7:30 AM - 9:30 AM) and evenings (5:00 PM - 8:00 PM).
- Congestion is commonly found in commercial hubs around Omalur and Salem due to local market activities and vehicular density.
- Roads near schools or industrial areas in Sangagiri can also experience short-term congestion.

## **4. Assessment of Road Quality and Infrastructure:**

- The road surface is generally well-maintained on highways and major roads, with regular signage and lighting.
- Potholes and wear may be present on smaller roads, especially in rural stretches or areas frequently hit by rainfall.
- Be cautious of narrow bridges and sharp turns, specifically on less frequented rural stretches between Desavilakku and Danishpet.

## **5. Suggestions for Alternative Routes for Emergencies:**

- In case of severe congestion or roadblock on the main route, consider using the NH 544 that bypasses Salem, or alternate state highways connecting through adjacent towns.
- Availability of side roads in rural stretches can be sparse; hence, plan and identify key exit routes before and through the journey for emergencies.

## **6. Local Regulations Affecting Hazardous Material Transport:**

- Transporting hazardous materials requires adherence to Tamil Nadu's guidelines and securing necessary permits.
- Time restrictions are sometimes imposed in residential and commercial zones, prohibiting heavy vehicle movement during peak hours.

## **7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials:**

- While specific databases should be used for detailed research, reports suggest occasional accidents involving vehicular collisions or spillages, typically attributed to human error or weather conditions.
- Regularly updated regulations and training have reduced frequent occurrences over recent years.

## **8. Environmental Considerations and Sensitive Areas:**

- The route traverses through regions of agricultural importance and some forest reserves, particularly near Danishpet.

- Reducing emissions and observing strict waste disposal protocols are vital to protecting these environments.

9. **Communication Coverage and Potential Dead Zones:**

- Most of the route provides adequate mobile network coverage; however, expect potential dead zones in sparsely populated rural areas, especially near forest fringes around Danishpet.
- Drivers should be equipped with offline navigation and emergency contact tools.

10. **Estimated Emergency Response Times:**

- Urban areas like Omalur and Salem offer faster emergency response (approximately 15-30 minutes) due to proximity to services.
- Rural sections may experience delayed response times, possibly extending to 45 minutes to an hour, given terrain and remoteness.

11. **Overall Summary of Risk Assessment:**

- The route presents manageable risks for transporting heavy and hazardous materials, provided adherence to weather advisories, road condition reports, and local transport regulations.
- Critical risks include monsoon-related issues, human factors during nighttime driving, and environmental protection compliance.
- Regular updates and en route communication are necessary to mitigate unforeseen disruptions.

Drivers and transport coordinators must follow these insights for safer and efficient transit along this crucial route in Tamil Nadu.

Risk Assessment - Turns

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 0  | Turn      | High       | 11.43961, 77.87341 | 15 KM/Hr    |
| 1  | Turn      | High       | 11.44024, 77.87528 | 15 KM/Hr    |
| 2  | Turn      | High       | 11.44898, 77.87410 | 15 KM/Hr    |
| 3  | Turn      | High       | 11.45352, 77.85698 | 15 KM/Hr    |
| 4  | Turn      | Medium     | 11.45847, 77.85140 | 30 KM/Hr    |
| 5  | Turn      | Medium     | 11.45898, 77.85135 | 30 KM/Hr    |
| 6  | Turn      | High       | 11.46303, 77.84945 | 15 KM/Hr    |
| 7  | Turn      | Medium     | 11.45543, 77.81740 | 30 KM/Hr    |
| 8  | Turn      | High       | 11.45631, 77.81668 | 15 KM/Hr    |
| 9  | Turn      | High       | 11.49383, 77.88442 | 15 KM/Hr    |
| 10 | Turn      | Medium     | 11.56083, 77.89921 | 30 KM/Hr    |
| 11 | Turn      | Medium     | 11.56180, 77.90019 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 12 | Turn       | Medium     | 11.58254, 77.90656 | 30 KM/Hr    |
| 13 | Turn       | Medium     | 11.63867, 77.94841 | 30 KM/Hr    |
| 14 | Turn       | Medium     | 11.64217, 77.95200 | 30 KM/Hr    |
| 15 | Turn       | Medium     | 11.64712, 77.95118 | 30 KM/Hr    |
| 16 | Turn       | High       | 11.67858, 77.96834 | 15 KM/Hr    |
| 17 | Turn       | High       | 11.70233, 77.97869 | 15 KM/Hr    |
| 18 | Turn       | Medium     | 11.71394, 77.99572 | 30 KM/Hr    |
| 19 | Turn       | Medium     | 11.71916, 78.01117 | 30 KM/Hr    |
| 20 | Turn       | Medium     | 11.72559, 78.01758 | 30 KM/Hr    |
| 21 | Turn       | Medium     | 11.73787, 78.03528 | 30 KM/Hr    |
| 22 | Turn       | High       | 11.73651, 78.03868 | 15 KM/Hr    |
| 23 | Turn       | Medium     | 11.74257, 78.04708 | 30 KM/Hr    |
| 24 | Blind Spot | Blind Spot | 11.87104, 78.08356 | 10 KM/Hr    |
| 25 | Turn       | Medium     | 11.87071, 78.08450 | 30 KM/Hr    |
| 26 | Turn       | Medium     | 11.87089, 78.08558 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.87249, 78.08707 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.86893, 78.12084 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 11.86491, 78.12535 | 30 KM/Hr    |
| 30 | Turn       | High       | 11.85820, 78.13580 | 15 KM/Hr    |
| 31 | Turn       | Medium     | 11.86652, 78.13879 | 30 KM/Hr    |
| 32 | Turn       | High       | 11.88411, 78.14478 | 15 KM/Hr    |
| 33 | Turn       | Medium     | 11.90581, 78.15292 | 30 KM/Hr    |
| 34 | Turn       | High       | 11.90760, 78.15205 | 15 KM/Hr    |
| 35 | Turn       | Medium     | 11.91226, 78.15594 | 30 KM/Hr    |
| 36 | Turn       | Medium     | 11.91312, 78.15616 | 30 KM/Hr    |
| 37 | Turn       | Medium     | 11.91519, 78.15837 | 30 KM/Hr    |
| 38 | Turn       | High       | 11.92454, 78.17073 | 15 KM/Hr    |
| 39 | Turn       | Medium     | 11.92820, 78.18040 | 30 KM/Hr    |
| 40 | Turn       | Medium     | 11.92797, 78.18165 | 30 KM/Hr    |
| 41 | Turn       | Medium     | 11.92890, 78.18331 | 30 KM/Hr    |
| 42 | Turn       | Medium     | 11.93569, 78.18711 | 30 KM/Hr    |
| 43 | Turn       | Medium     | 11.95116, 78.20647 | 30 KM/Hr    |
| 44 | Turn       | Medium     | 11.95200, 78.20703 | 30 KM/Hr    |
| 45 | Turn       | High       | 11.95258, 78.20848 | 15 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 46 | Turn       | Medium     | 11.95173, 78.21017 | 30 KM/Hr    |
| 47 | Turn       | High       | 11.95221, 78.21173 | 15 KM/Hr    |
| 48 | Turn       | Medium     | 11.95165, 78.21276 | 30 KM/Hr    |
| 49 | Turn       | Medium     | 11.95228, 78.21413 | 30 KM/Hr    |
| 50 | Turn       | Medium     | 11.95220, 78.21522 | 30 KM/Hr    |
| 51 | Turn       | Medium     | 11.95288, 78.21603 | 30 KM/Hr    |
| 52 | Turn       | Medium     | 11.95264, 78.22467 | 30 KM/Hr    |
| 53 | Turn       | High       | 11.95178, 78.22600 | 15 KM/Hr    |
| 54 | Turn       | Medium     | 11.95843, 78.23409 | 30 KM/Hr    |
| 55 | Turn       | Medium     | 11.96188, 78.23441 | 30 KM/Hr    |
| 56 | Turn       | High       | 11.96666, 78.23282 | 15 KM/Hr    |
| 57 | Blind Spot | Blind Spot | 11.96699, 78.23204 | 10 KM/Hr    |
| 58 | Turn       | High       | 11.97325, 78.23714 | 15 KM/Hr    |
| 59 | Turn       | High       | 11.97765, 78.23556 | 15 KM/Hr    |
| 60 | Turn       | High       | 11.98243, 78.24308 | 15 KM/Hr    |
| 61 | Turn       | High       | 11.98206, 78.24332 | 15 KM/Hr    |
| 62 | Turn       | Medium     | 11.99307, 78.25808 | 30 KM/Hr    |

## Emergency Locations

|    | type     | name                              | coordinates            | speed_limit | risk_level |
|----|----------|-----------------------------------|------------------------|-------------|------------|
| 1  | hospital | Sri Shanthi Hospital              | 11.569272, 77.901937   | 30 km/h     | Medium     |
| 2  | hospital | Omalur Government Hospital        | 11.74161, 78.044296    | 30 km/h     | Medium     |
| 3  | clinic   | R.R. Nursing Home                 | 11.7394129, 78.0439861 | 30 km/h     | Medium     |
| 4  | hospital | Sri Hari Hospital                 | 11.7396726, 78.0440701 | 30 km/h     | Medium     |
| 5  | hospital | Sugumar Hospital                  | 11.7396031, 78.0426955 | 30 km/h     | Medium     |
| 6  | hospital | Sri Sugam Hospital                | 11.7429821, 78.0461039 | 30 km/h     | Medium     |
| 7  | hospital | Saravana Hospital                 | 11.7409303, 78.0458308 | 30 km/h     | Medium     |
| 8  | hospital | Arun Hospital                     | 11.7429892, 78.0466881 | 30 km/h     | Medium     |
| 9  | clinic   | Saratha Clinic                    | 11.7422818, 78.0458175 | 30 km/h     | Medium     |
| 11 | clinic   | Sri Meenakshi Nursing Home, Salem | 11.7449162, 78.0474637 | 30 km/h     | Medium     |
| 12 | clinic   | Divyam Child Care Clinic          | 11.7447135, 78.0473858 | 30 km/h     | Medium     |

|    | type     | name                               | coordinates            | speed_limit | risk_level |
|----|----------|------------------------------------|------------------------|-------------|------------|
| 13 | clinic   | Momota Clinic                      | 11.744982, 78.047397   | 30 km/h     | Medium     |
| 14 | clinic   | Gomathi Clinic                     | 11.7445293, 78.0470401 | 30 km/h     | Medium     |
| 15 | hospital | Vinayaka Child Hospital            | 11.745812, 78.0480225  | 30 km/h     | Medium     |
| 16 | hospital | Dr. G.R.Thilakam Hospital          | 11.746097, 78.04769    | 30 km/h     | Medium     |
| 17 | hospital | Thulasi Hospital                   | 11.7461516, 78.0469984 | 30 km/h     | Medium     |
| 18 | hospital | Amirthalingam Hospital             | 11.7456708, 78.0470548 | 30 km/h     | Medium     |
| 19 | hospital | Children Hospital                  | 11.7458377, 78.0480205 | 30 km/h     | Medium     |
| 20 | clinic   | Shri Ramana Poly Clinic            | 11.7477122, 78.0479293 | 30 km/h     | Medium     |
| 21 | hospital | Dr. Nagaraj Hospital               | 11.7464035, 78.0472401 | 30 km/h     | Medium     |
| 22 | clinic   | Manimuthu Skin Clinic              | 11.7521189, 78.0484058 | 30 km/h     | Medium     |
| 23 | hospital | St. Joseph's Speciality Hospital   | 11.761611, 78.0522344  | 30 km/h     | Medium     |
| 24 | hospital | Anugraha Multi Speciality Hospital | 11.773549, 78.0546408  | 30 km/h     | Medium     |
| 26 | hospital | L.R. Hospital                      | 11.8703354, 78.0844554 | 30 km/h     | Medium     |
| 27 | clinic   | JSA Clinic                         | 11.8700619, 78.0844982 | 30 km/h     | Medium     |
| 28 | clinic   | Kamala Clinic                      | 11.8723673, 78.0874195 | 30 km/h     | Medium     |
| 29 | hospital | Kamala Hospital                    | 11.8722588, 78.0899189 | 30 km/h     | Medium     |
| 30 | hospital | GJB Hospital                       | 11.8713998, 78.0980288 | 30 km/h     | Medium     |
| 31 | clinic   | Pranav Clinic                      | 11.8712846, 78.0998167 | 30 km/h     | Medium     |
| 32 | hospital | Bethel Hospital                    | 11.8699472, 78.1388598 | 30 km/h     | Medium     |
| 33 | hospital | Amirtham Hospital                  | 11.981994, 78.248038   | 30 km/h     | Medium     |
| 34 | hospital | Grs Hospital                       | 11.9824068, 78.2481763 | 30 km/h     | Medium     |

### Crowded Spots

|    | type    | name                                   | coordinates            | speed_limit | risk_level |
|----|---------|----------------------------------------|------------------------|-------------|------------|
| 0  | school  | KRP Matric. Hr. Sec School             | 11.4546193, 77.8142445 | 30 km/h     | Medium     |
| 10 | school  | Fathima Hr Sec School                  | 11.7410508, 78.0479272 | 30 km/h     | Medium     |
| 25 | college | Dhirajlal Gandhi College of Technology | 11.7968014, 78.0562095 | 30 km/h     | Medium     |

### Route Photos of Risky Spots





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44024, 77.87528



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44898, 77.87410





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45352, 77.85698



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45847, 77.85140



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45898, 77.85135





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.46303, 77.84945



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45543, 77.81740





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45631, 77.81668



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49383, 77.88442



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.56083, 77.89921





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.56180, 77.90019



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.58254, 77.90656





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.63867, 77.94841



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64217, 77.95200



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64712, 77.95118





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.67858, 77.96834



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.70233, 77.97869





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.71394, 77.99572



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.71916, 78.01117



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.72559, 78.01758





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.73787, 78.03528



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.73651, 78.03868





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.74257, 78.04708



**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.87104, 78.08356



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.87071, 78.08450





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.87089, 78.08558



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.87249, 78.08707





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.86893, 78.12084



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95116, 78.20647



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.97325, 78.23714





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.97765, 78.23556



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.98243, 78.24308





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.98206, 78.24332



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.99307, 78.25808

---